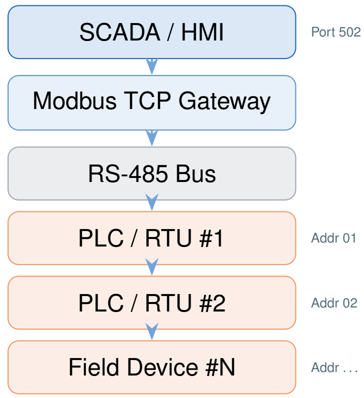
Modbus Protocol

12 Chapters · 500+ Questions · PDF Included

Study Guide · Industrial Automation Series · SCADA Hub Training Platform



WHAT THIS COURSE TEACHES

Modbus is the first protocol every SCADA engineer encounters and the last one they fully understand. Written in 1979 and still running millions of devices in active service, it underpins more industrial infrastructure than any protocol introduced since. This course teaches Modbus from first principles through field-level troubleshooting — not as an academic exercise, but as the engineering foundation you need before every other protocol makes sense.

CHAPTER SEQUENCE

1. **Introduction to Modbus** — History, RTU vs. ASCII vs. TCP, the master/slave model, RS-485 specifications.
2. **Data Model** — The four register tables, 0- vs. 1-based addressing, data types, packed coil transmission.
3. **Function Codes** — FC01–FC23, exception codes, the request/response pair structure.
4. **RTU Framing** — Binary encoding, CRC-16 calculation, timing-based frame delimitation.
5. **ASCII Mode** — LRC checksum, colon/CRLF delimiters, tradeoffs vs. RTU.
6. **Modbus TCP** — MBAP header, port 502, transaction IDs, gateway behavior.
7. **RS-485 Physical Layer** — Differential signaling, termination, cable length vs. baud rate.
8. **Data Types & Scaling** — 32-bit floats, word order, engineering unit conversion.
9. **Security** — Attack surface analysis, network segmentation strategy, protocol-aware firewall rules.
10. **Troubleshooting** — Exception code diagnosis, timeout analysis, common commissioning failures.
11. **Advanced Topics** — Multi-drop timing, gateway configuration, Modbus over satellite/radio.
12. **Protocol Lab** — Live exercises: frame construction, CRC verification, Wireshark capture analysis.

THE STORY

Modbus is the entry point. Engineers who skip it — or who learned it "on the job" without structure — carry gaps that compound as they move up the stack. An engineer who doesn't understand register addressing will struggle with every integration they touch. An engineer who doesn't understand RTU framing cannot diagnose a serial fault without guessing.
This course closes those gaps in sequence: physical layer → framing → function codes → TCP encapsulation → security → lab. By the end, the engineer can read a raw Modbus frame in hex and tell you what register is being read, from which slave, and whether the response is correct — without opening a manual.

Certification Relevance

- **ISA CCST** — Modbus is explicitly tested in the Instrumentation & Control systems domain.
- **GICSP (GIAC)** — ICS protocol knowledge; Modbus is a core topic in all vendor prep materials.
- **CompTIA CySA+** — ICS/OT protocol awareness tested at associate level.

Next Steps (Certs)

1. Complete this course + Wireshark guide for hands-on lab skills.
2. Study ISA-18.2 and ISA-99 (IEC 62443) standards.
3. Register at `isa.org` for CCST; at `giac.org` for GICSP.
4. Schedule exam within 90 days of course completion.

12 Chapte **500+** Questions **3** Levels